

# Exploring the socioeconomic and ecological consequences of cash crop cultivation for policy implications

Jing Li, Zeli Zhang\*, Xianfeng Jin\*, Jiaquan Chen, Shaojia Zhang, Zong He, Sheng Li, Zhiming He, Haipeng Zhang, He Xiao

Chongqing Geomatics Center, Chongqing, 401121, China



## ARTICLE INFO

### Keywords:

Landscape pattern  
Land use change  
Socioeconomic indicators  
Ecosystem service value  
Cash crop cultivation

## ABSTRACT

Cash crops have kept expanding at an accelerating rate across the globe during the last decades. It therefore requires elaborate efforts to examine the socioeconomic and ecological consequences of cash crop cultivation. With a case of the Hangzhou region in subtropical China, this paper investigated the dynamic patterns of four cash crop types (tea, fruit, mulberry and nursery) at town level by using aerial photos; and then quantified the subsequent socioeconomic and ecological consequences using spatial regression. In particular, the socioeconomic impacts were examined based on a set of socioeconomic indicators and the ecological consequences were described by ecosystem service values (ESV) and landscape pattern changes. Results indicated that the economic benefits of cash crop cultivation were evident, including raising household income, boosting rural economy, increasing fiscal revenues, and attracting foreign investment. Cash crop cultivation generally yielded positive social impacts (welfare promotion, infrastructure improvement and job creation), but the impacts also varied with crop types. Cash crop cultivation not only increased landscape fragmentation, isolation and irregularity, but also decreased the dominance, connectivity and aggregation of forest and farmland. Specifically, forest was more subjected to tea and fruit expansion, while farmland was more vulnerable to mulberry and nursery expansion. A significant negative relationship was identified between ESV changes and cash crop expansion. It implied that cash crop cultivation would impair the capacity of ecosystems to deliver services. Our study demonstrated an applicable framework to identify the essential indicators for land use policy makers to monitor the socioeconomic and ecological consequences of cash crop cultivation.

## 1. Introduction

Global demand for cash crops has kept increasing during the recent past (Delpuch and Leblois, 2014). Cash crop cultivation, including the rubber, commercial fruit, palm oil, tea and nursery, has become an expanding global phenomenon, especially in tropical and subtropical countries (Carlson et al., 2012; Castiblanco et al., 2013; Gatto et al., 2015; Godone et al., 2014; Qiu, 2009; Xiao et al., 2015; Ziegler et al., 2009). However, the contribution of cash crop cultivation to social welfare, economic development and ecological conservation has sparked a controversial debate (Castiblanco et al., 2015; German et al., 2011). For one thing, cash crop cultivation is regarded as a source of export and a critical contributor to economic growth for the producer regions, and has accelerated local economies integration into global economies (Dal Belo Leite et al., 2015; GLOBAL, 2011; Rist et al., 2010; Su et al., 2016, 2017a,b; World Bank, 2010; Obidzinski et al., 2012; Zhang et al., 2014). For another, cash crop cultivation is criticized for

increasing negative social and environmental externalities, since it promotes competition in natural resources with traditional staple food crops, puts pressures on food inflation, raises conflicts over land rights (Barkmann et al., 2010; Ewing and Msangi, 2009; McCarthy, 2010; Tilman et al., 2009). It is notable that cash crop expansion usually occurs with deforestation, and this land use conversion can lead to various ecological problems (e.g., soil erosion, biodiversity loss, hydrological turbulence, raw materials reduction, and decreasing carbon stock) (Cha, 2005; Carlson et al., 2012; Fitzherbert et al., 2008; Petsri et al., 2013; Yi et al., 2014a). Specifically, previous studies pointed that bird diversity and abundance decreased along with increasing revenues from oil palm plantations; soil organic carbon stocks had decreased by converting forest to rubber. In addition, while the enlarging cash crop cultivation strengthens the local finance stability, its impact on socioeconomic welfare of households in the long run remains controversial (Castiblanco et al., 2015; German et al., 2011). Amid this controversy, it requires elaborate efforts to examine the

\* Corresponding authors at: Ranjiaba Rd, Chongqing, China.

E-mail addresses: [410154768@qq.com](mailto:410154768@qq.com) (Z. Zhang), [king@dl023.net](mailto:king@dl023.net) (X. Jin).

socioeconomic and ecological consequences of cash crop cultivation.

It is not an easy endeavor to establish a thorough examination of the consequences of cash crop cultivation. First, accurate information of the spatiotemporal patterns of cash crop cultivation is unavailable for most places. Second, fine level (e.g., village and town) socioeconomic data-base are scarce in most developing countries. Household survey always relies on subjective indicators, which incorporates bias in sample choice and impedes the assessment of impacts at broad level (e.g., city and state) (Castiblanco et al., 2015). Third, it is rather difficult to quantify the dynamic changes of ecosystem functions and structures, since long-term observation data are always absent and intensive field trips would generate very high costs (Mairota et al., 2013; Sowińska-Świerkosz and Soszyński, 2014; Su et al., 2014a).

Multitemporal remotely sensed images provide essential data source to monitor land use and cover changes in space and time. Recent literature has demonstrated that remote sensing was a practical and promising tool to obtain the spatiotemporal information of cash crops (Castiblanco et al., 2013; Dong et al., 2013; Godone et al., 2014; Li and Fox, 2012; Xiao et al., 2015). Ecological functions and processes, evidenced by a number of experiments, are significantly influenced by landscape patterns (Girvetz et al., 2008; Uemaa et al., 2013). Scholars generally agree that ecological consequence of anthropogenic activity can be indirectly quantified by analyzing landscape pattern changes (Fernandes et al., 2011; Sowińska-Świerkosz and Soszyński, 2014; Su et al., 2012; Weng, 2007). Landscape ecological approach, compared to field survey, permits a more rapid assessment of eco-environmental quality at regional scale (Fernandes et al., 2011). Besides, it provides quantitative and comparative figures to document the dynamic changes of eco-environmental quality (Su et al., 2014a). However, brief description of landscape pattern changes may not offer a full-view of the ecological consequences of anthropogenic activity, since the economic values of ecosystems are not revealed (Su et al., 2012; Wainger et al., 2010). Ecosystem services refer to the benefits provided by ecosystems to support and maintain the life and living of human beings (Millennium Ecosystem Assessment, 2003). Evaluation of the ecosystem service values (ESV) has shown promising potential to combine the economic outcomes with ecological processes (Wainger et al., 2010). Su et al. (2012) argued that monetary evaluation of ecosystem service values (ESV) and characterization of landscape patterns should be both applied to quantify the ecological consequences of anthropogenic activity. However, rather few studies have been done to investigate regional ESV and landscape pattern changes in relation to cash crop cultivations. In addition, the socioeconomic and ecological consequences of cash crop cultivation have been rarely examined simultaneously in the literature.

Considering the above limitations, this paper attempted to quantify the socioeconomic and ecological consequences of cash crop cultivation. Data at town level were collected for the Hangzhou region, a typical cash crop production place in subtropical China. Our specific objectives are to: (1) investigate the dynamic patterns of cash crop cultivation at town level within Hangzhou region; (2) examine the socioeconomic impacts of cash crop cultivation; and (3) analyze the ecological consequence of cash crop cultivation.

## 2. Materials and method

### 2.1. Study area

The Hangzhou region is situated in eastern coastal part of Zhejiang Province (Fig. 1), one of the most developed and populous administrations in China. Covering about 17,000 km<sup>2</sup>, it is hilly in the south-west and plain in the northeast. The subtropical monsoon climate controls the Hangzhou region, with plenty of sunshine (annual 1641 h) and rainfall (1274 mm). In addition, the red soils are loamy and thick and river networks are dense and connected. Thereby it provides a perfect farming environment to grow cash crops. A diversity of cash

crops with high yield and quality grow in Hangzhou region and the most famous ones include fruit, tea, nursery, and mulberry (Su et al., 2016). Local specialty cash crops of Hangzhou enjoy a high reputation and show obvious price advantage in markets. In pursuit of high income, the farmers have showed higher and higher enthusiasm towards cash crop cultivation. The cultivated area of cash crops has expanded to a large extent during the past decade (Su et al., 2016). It is argued that Hangzhou region should be a typical case of cash crop expansion and should be ideal for examining the socioeconomic and ecological consequences of cash crop cultivation.

### 2.2. Digital land use data

The 2.5 m resolution color-infrared aerial photos, provided by the Su et al. (2016) were employed to interpret land use information. These photos were imported to ArcGIS 10.2 with WGS84 geographic coordinates and then visually interpreted at 1: 50,000 photogrammetric scale. The interpreters were selected from the local experts and all were familiar with the land use patterns in Hangzhou. In particular, considering the actual land use characteristics in Hangzhou, we interpreted four cash crop types (tea, orchard (fruit), mulberry and nursery). In order to facilitate the metric analysis and ESV calculation, six other land use types (farmland, forest, artificial ponds, natural water bodies, built-ups and fallow land) were also interpreted. In order to evaluate the accuracy, we used field survey data and the number of reference points amounted to 235 points for 2004 and 298 points for 2014. The land use distributions were presented in Fig. 2. The interpreted cash crop information was joined with the digital administrative map (1: 50,000) at town level (Fig. 1). Then, the area of the four cash crop types was summed for each town (146 in total). It should be mentioned that administration in China is divided into four levels, including province, city, town (district) and village (community). In most cases, comprehensive socioeconomic statistical data are available at town level rather than village level. Analysis was therefore conducted at town level in this study.

### 2.3. Selection of landscape metrics

Metric analysis permits the numerical and comparative description of landscape patterns. A large amount of metrics has been developed to depict the landscape characteristics from different aspects, including size, edge, connectivity, shape and diversity (Girvetz et al., 2008; Uemaa et al., 2013). However, landscape metrics usually present high redundancy in most cases (Sowińska-Świerkosz and Soszyński, 2014). It is therefore suggested to choose a set of metrics rather than to employ all the metrics. The selection of landscape metrics was based on four criteria: (1) comparability with previous studies (Kromroy et al., 2007; Leitão and Ahern, 2002; Pôças et al., 2011; Solon, 2009; Su et al., 2011; Weng, 2007); (2) ability to reflect ecological conditions (Ribeiro and Lovett, 2009); (3) low redundancy among landscape metrics by performing a multicollinearity test, calculating metrics at eco-regional scale (Leitão and Ahern, 2002); and (4) ability to indicate characteristics of landscape patterns of the study area.

Metrics selection in this study followed the three-step procedure demonstrated in Su et al. (2014a). First, literature review was based on to select a set of metrics (30 landscape level and 38 class level), which represented the landscape pattern characteristics from various aspects (e.g., density, edge, area, isolation, shape, diversity and contagion). Second, after normalization and standardization, these indices were subjected to Pearson's correlation analysis. When one pair of metrics were found to be highly correlated ( $|r| \geq 0.9$ ), one of them was abandoned. Third, the remaining indices were further subjected to principal component analysis with varimax rotation. When one metric in the extracted component presented high loadings ( $> 0.75$ ), it was kept. Finally, five landscape level metrics (patch density (PD), landscape shape index (LSI), connectance index (CONNECT), split index (SPLIT)



Fig. 1. Location and administrative divisions at town level of the Hangzhou region, China.

and Shannon's diversity index (SHDI)), and five class level metrics (total area (TA), edge density (ED), mean Euclidean nearest neighbour distance (ENND), area weighted shape index (AWSI), and aggregation index (AI)) were selected. It should be mentioned that class level metric calculation was only performed for farmland and forest, given that cash crop expands by replacing farmland and forest (Xiao et al., 2015). All the metrics were facilitated by using the Patch Analyst 4.0 (Rempel, 2008). Ecological significance of the selected metrics was summarized in Appendix A.

#### 2.4. Evaluation of ecosystem service values

Land use proxy based approach (Costanza et al., 1997) was employed to evaluate the ESV. This approach has received criticism for producing high variations and uncertainty in magnitude estimations. However, this method is insensitive for estimating directional changes, since the bias can be largely reduced in temporal trend analysis (Su et al., 2014b). Previous studies have demonstrated the land use proxy based approach was effective in capturing the ESV changes in the study area (Su et al., 2012; Wu et al., 2013).

The equation for calculating ESV was described as follows.

$$ESV = \sum (A_i \times C_i) \quad (1)$$

where  $A_i$  is area of land use type  $i$  and  $C_i$  is the value coefficient for land use type  $i$ .

We referred to Su et al. (2012) and (2014b) for the coefficients for farmland, forest, artificial ponds, natural water bodies, built-ups and

fallow land (Table 1). We also referred to Zhang (2012) to calculate the coefficients for the tea, orchard, mulberry and nursery (Table 1). These coefficients were proven to be reliable by sensitivity test.

#### 2.5. Selection of socioeconomic indicators

Selection of socioeconomic indicators involved three steps. First, a set of indicators were proposed following three principles: (1) they can reflect the social welfare, economic development and life quality of households; (2) previous literature demonstrated that they had potential correlation with cash crop cultivation; and (3) data for generating these indicators were available. Second, these indicators were subjected to Pearson's correlation analysis and principal component analysis, the procedure of which was identical to that for metric selection in Section 2.3. Third, expert panel evaluation was performed to further reduce the remaining indicators. Twelve experts engaged in the evaluation, among whom four were ecologists, four were geographers, two were sociologists and two were from the local land management bureau. These experts were asked to assign a score ranging from 1 to 5, considering the appropriateness of the indices. T test showed that scores assigned by the experts had no significant differences ( $P = 0.11$ ). Then, the average score for each index was summed by assigning equal weight to the ten experts. During the expert panel discussion, one index was criticized for incorporating shortcomings and it was abandoned. One index was revised based on the comments of the experts. The indices whose scores were above the average level were kept. The final selected socioeconomic indices were listed in Table 2. All the original data for these

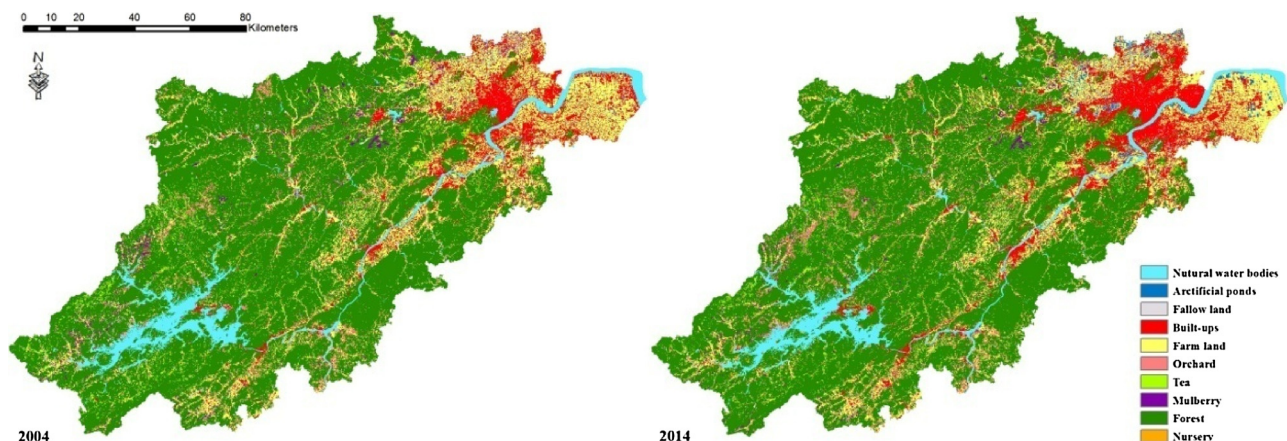


Fig. 2. Land use in 2004 and 2014 within Hangzhou region, China.

**Table 1**

Coefficients of ecosystem service values for different land use types in Hangzhou region, China (unit: RMB Yuan/ha).

|                      | Gas regulation | Climateregulation | Waterreservation | Soil formationand protection | Biodiversityconservation |
|----------------------|----------------|-------------------|------------------|------------------------------|--------------------------|
| Farmland             | 885.0          | 1575.2            | 1062.1           | 2584.0                       | 1256.4                   |
| Tea                  | 547.9          | 1835.6            | 554.3            | 1033.7                       | 719.1                    |
| Orchard              | 368.4          | 1692.4            | 795.3            | 1278.9                       | 897.8                    |
| Mulberry             | 758.8          | 1987.5            | 1125.8           | 1847.6                       | 794.5                    |
| Nursery              | 1579.6         | 2075.6            | 1986.4           | 2057.9                       | 1054.7                   |
| Forest               | 3097.0         | 2389.1            | 2831.5           | 3450.9                       | 2884.6                   |
| Natural water bodies | 0              | 407.0             | 118033.2         | 8.8                          | 2203.3                   |
| Artificial ponds     | 0              | 105.4             | 1014.5           | 3.5                          | 574.6                    |
| Fallow land          | 0              | 0                 | 0                | 0                            | 0                        |
| Built-ups            | 0              | 0                 | 0                | 0                            | 0                        |

indicators were provided by the local government.

## 2.6. Spatial regression

In order to quantify the socioeconomic impacts of cash crop cultivation, we analyzed the relationships between changes of socioeconomic indicators and cash crop expansion. The ecological consequences were quantified by analyzing landscape pattern and ESV changes in relation to cash crop expansion. Landscape patterns and socioeconomic development always present significant autocorrelation. It implies that the traditional ordinary least square regression (OLS) is limited in interpreting the consequences of cash crop cultivation. The spatial regression was therefore employed to solve this problem. In particular, spatial dependency is incorporated into the spatial regression in the form of lag or error (Anselin, 1988; LeSage and Pace, 2009). Eqs. (2) and (3) formulate the spatial lag and error regression, respectively.

$$Y = \beta W_Y + \alpha X + \eta + \varepsilon \quad (2)$$

$$Y = \alpha X + \eta + \varepsilon (\varepsilon = \beta W_\varepsilon + \gamma) \quad (3)$$

where  $X$  is the independent variable and  $Y$  is the dependent variable;  $\alpha$  and  $\beta$  are the coefficients;  $\gamma$  and  $\eta$  are scalar parameters;  $\varepsilon$  is the error term;  $W_Y$  is the spatial matrix for the dependent variable and  $W_\varepsilon$  is the spatial matrix for error term.

Before performing the spatial regression using the GeoDa 0.9.5-i (Beta) software (Anselin et al., 2006), we normalized and standardized all the parameters. The robust Lagrange multiplier tests were used for specifying the lag or error model (LeSage and Pace, 2009). In particular,

bivariate regression was applied, in order to compare the impacts of different cash crop types. Areal changes of cash crops were the explanation variables. Changes of socioeconomic indicators, landscape metrics and ESV (Eq. (4)) were the dependent variables. Towns, where cash crops were not cultivated, were excluded into analysis. Data of the socioeconomic indicators for 16 towns were missing, and these towns were also excluded during analysis. Extremes and outliers were also excluded into regression, in order to enhance the performance of statistical procedures.

$$C_x = \frac{x_{i+t} - x_i}{x_i} \times 100\% \quad (4)$$

where  $C_x$  is the changes of variable  $x$  between time  $i$  and time  $i + t$ ;  $x_i$  is value of variable  $x$  in time  $i$ ;  $x_{i+t}$  is the value of variable  $x$  in time  $i + t$ .

## 3. Results

### 3.1. General descriptions

Patterns of cash crop expansion within Hangzhou region were shown in Fig. 3. Majority of the towns experienced dramatic cash crop expansion from 2004 to 2014. Specifically, tea, fruit and mulberry expansion was more substantial in the southwestern towns. Greater amount of nursery expansion was observed in the northwestern towns. As shown in Fig. 4, it can be seen that landscape pattern changes presented great heterogeneity. However, towns presented identical trend: decreases in CONNECT and increases in PD, LSI, SPLIT and SHDI. It suggested that landscape patterns became less connected and more fragmented and irregular. Considering the class-level landscape pattern

**Table 2**

Selected indicators describing the potential socioeconomic impacts of cash crop cultivation in Hangzhou region, China.

| Dimension            | Indicator                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Social welfare       | Basic welfare index               | An integrated index developed by the local government. Ten indices (Number of doctors per 10,000 persons, Number of welfare houses per 10,000 persons, Number of hospitals per 10,000 persons, Number of teachers per 100 students, Number of banks and credit cooperatives per 10,000 persons, Number of kindergartens per 100 children, Number of bus stations per 10,000 persons, Number of waste disposal stations per 10,000 persons, Number of tank stations per 10,000 persons and Number of marketplaces per 10,000 persons) were standardized and integrated by principle component analysis method. |
|                      | Employment rate                   | Percentage of employed population                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | Infrastructure construction index | Ratio between infrastructure fixed assets investment and total fixed assets investment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Economic development | Average rural income              | Average income of farmers from all sources including cash and physical assets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | Total income of rural economy     | Town operating income that can be used to compensate for expenses for the current year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | Fiscal income index               | Ratio between fiscal revenue and fiscal expenditure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                      | Foreign investment                | Total value of the foreign investment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Life quality         | Basic needs index                 | An integrated index developed by the local government. Four indices (Percentage of household having flush toilet, Percentage of household having tap water, Percentage of household having fuel gas, and Percentage of household having fixed-line telephone) were standardized and integrated by entropy weight method.                                                                                                                                                                                                                                                                                      |
|                      | Advanced needs index              | An integrated index developed by the local government. Four indices (Percentage of household having cell phone, Percentage of household access to internet, Percentage of household having motorcycle and Percentage of household having private car) were standardized and integrated by entropy weight method.                                                                                                                                                                                                                                                                                              |
|                      | Emigration rate                   | Percentage of emigration population                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



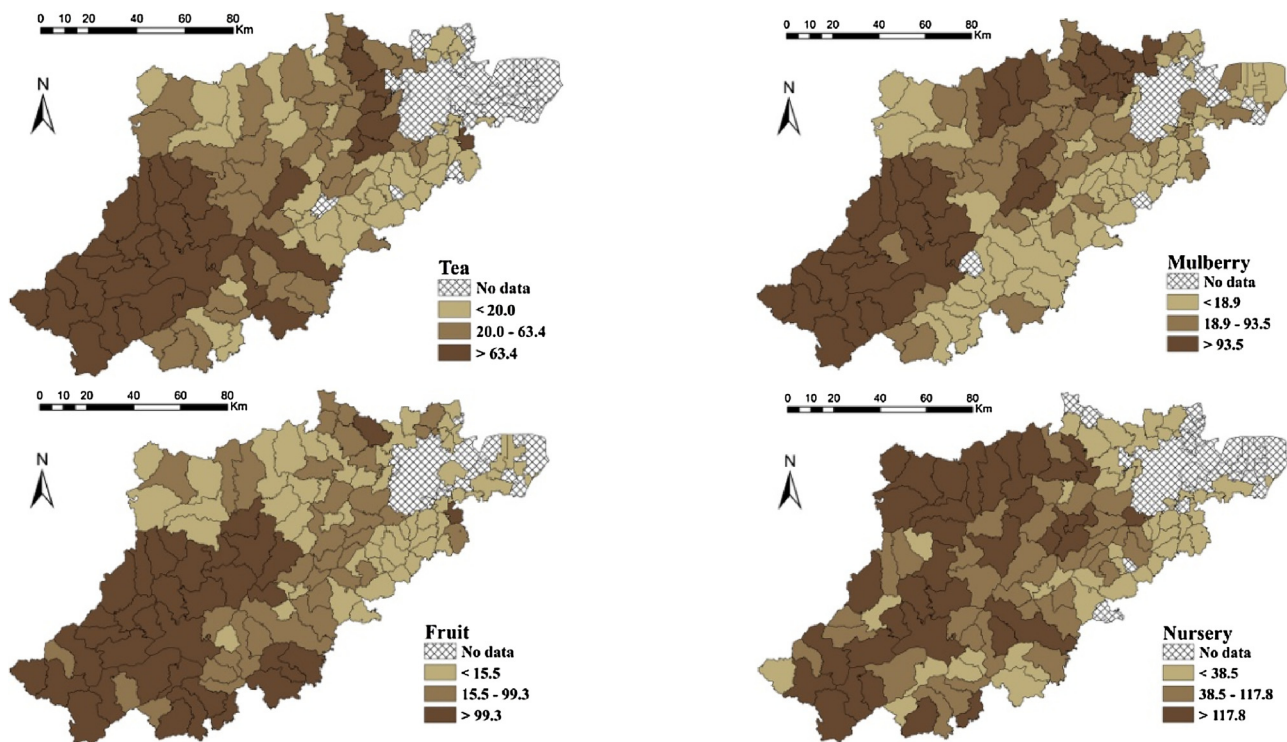


Fig. 3. Expansion of different cash crop types at town level from 2004 to 2014 within Hangzhou region, China (unit: ha).

changes (Fig. 5), it can be inferred that forest and farmland were less dominated, aggregated and connected but more irregular. As shown in Fig. 6, the towns within Hangzhou region generally presented a declining tendency of ESV changes. It implied that the capacity of ecosystems to deliver services for human beings should be impaired. In particular, changes of ESV were more substantial in the southwestern towns. Fig. 7 displayed the spatial patterns of socioeconomic indicators, which presented great heterogeneity. However, towns presented identical trend: increases in all the socioeconomic indicators. It implied that rural economy developed rapidly in these towns and social welfare as well as life quality of households were promoted during the study period.

### 3.2. Relationships between changes of landscape metrics and cash crop expansion

Table 3 showed the relationships between changes of landscape-level metrics and cash crop expansion. The estimated coefficients and model explanatory ability varied with cash crop types; however, the obtained relationships exhibited identical nature. Specifically, increases of the four cash crop types were negatively associated with changes of CONNECT but positively correlated with changes of PD, LSI, SPLIT and SHDI. It suggested that cash crop expansion could significantly lead to landscape fragmentation, isolation and irregularity.

At the class-level, expansion of the four cash crop types also had

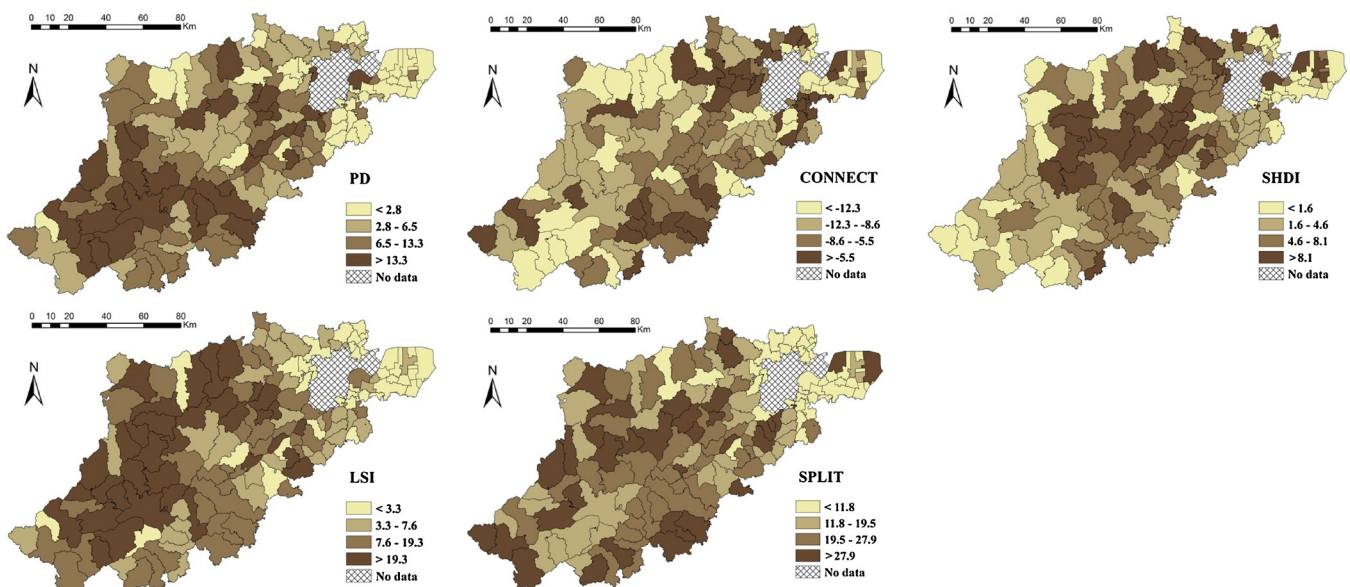


Fig. 4. Changes of landscape-level metrics at town level from 2004 to 2014 within Hangzhou region, China (unit: %): patch density (PD), landscape shape index (LSI), connectance index (CONNECT), split index (SPLIT) and Shannon's diversity index (SHDI).

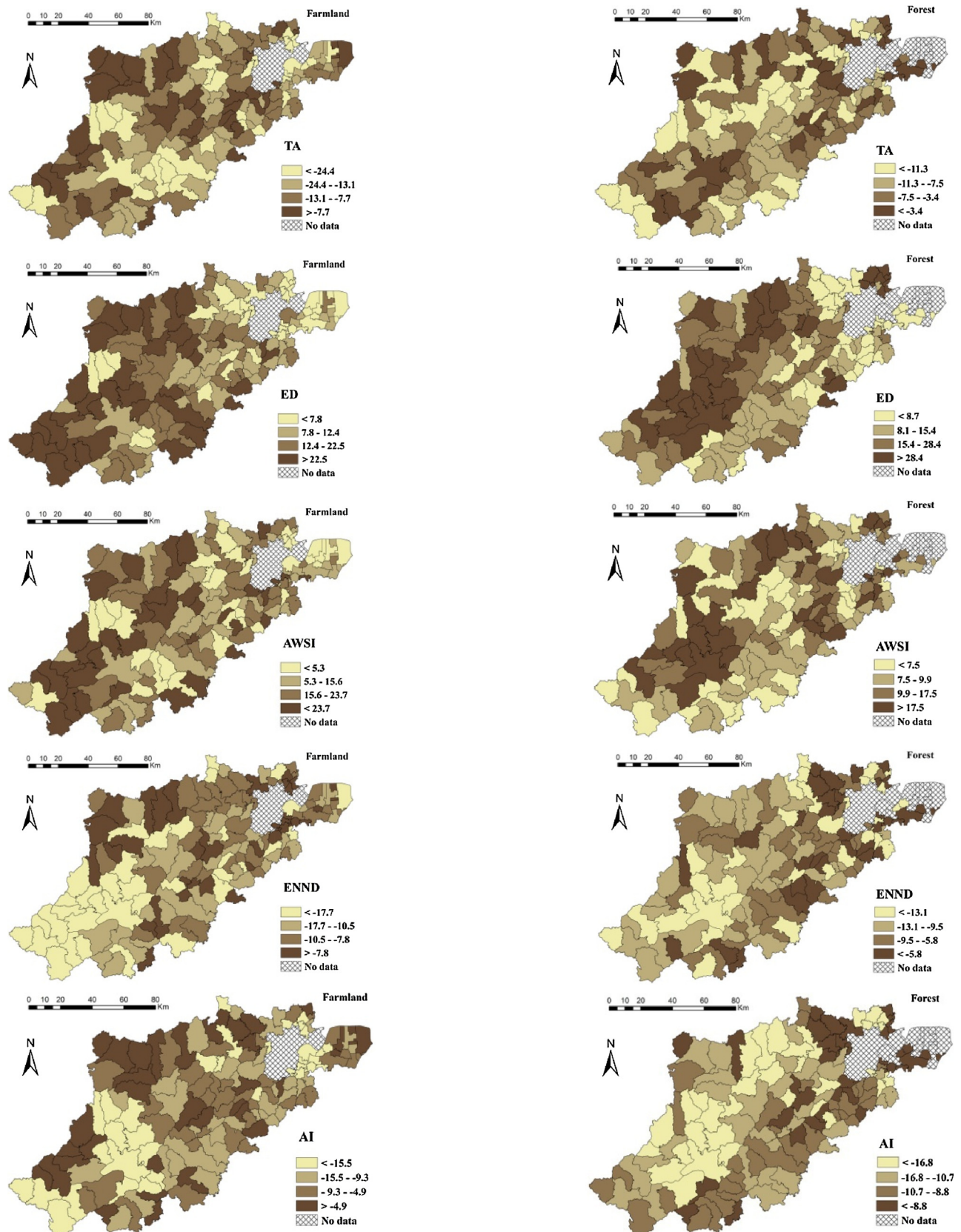


Fig. 5. Change rate of landscape-level metrics at town level from 2004 to 2014 within Hangzhou region, China (unit: %): total area (TA), edge density (ED), mean Euclidean nearest neighbour distance (ENND), area weighted shape index (AWSI), and aggregation index (AI).



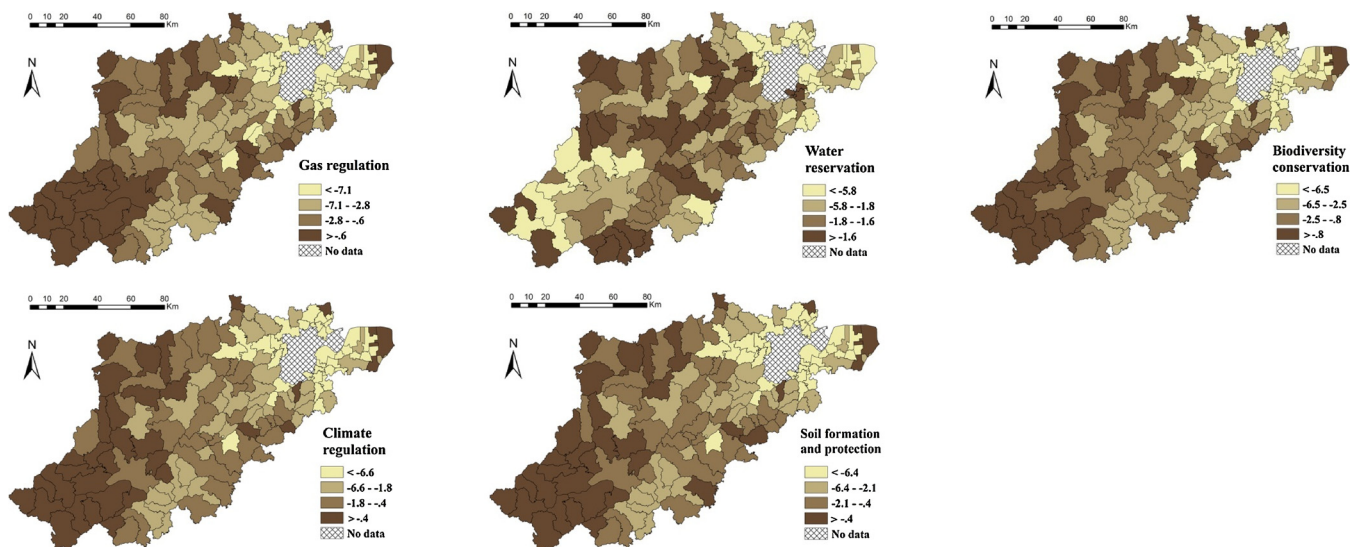


Fig. 6. Changes of values for different ecosystem services at town level from 2004 to 2014 within Hangzhou region, China (unit: %).

significant impacts on landscape pattern changes (Table 4). Specifically, for the forest landscape pattern changes, the influences of the tea and fruit expansion were stronger than those of mulberry and nursery expansion, given the higher explained variance. For the farmland landscape pattern changes, the influence of tea expansion was obviously lower than that of the other cash crop types. The negative coefficients for TA, AI and ENND denoted that cash crop expansion would decrease the dominance, connectivity and aggregation of forest and farmland. The positive coefficients for ED and AWSI signified that forest and farmland patches would become more instable and irregular in response to cash crop expansion.

### 3.3. Relationships between changes of ESV and cash crop expansion

Table 5 demonstrated the relationships between ESV changes and cash crop expansion. In particular, changes of the values for the five ecosystem services were all negatively correlated with the expansion of the four cash crop types. It implied that cash crop expansion could result in the declines of ESV. Comparing the variance explained by different cash crop types, it can be inferred that ESV changes were more sensitive to tea and fruit expansion.

### 3.4. Relationships between changes of socioeconomic indicators and cash crop expansion

Relationships between changes of socioeconomic indicators and cash crop expansion were shown in Table 6. Among the three social welfare indicators, the basic welfare index and the infrastructure construction index had positive correlations with the expansion of the four cash crop types. It denoted that cash crop cultivation could promote social welfare in the rural areas. The employment rate was significantly correlated with tea, fruit and mulberry expansion. It implied that tea, fruit and mulberry plantations would possibly increase the local job opportunities. The four economic development indicators presented positive relationships with the expansion of the four cash crop types. It revealed that cash crop cultivation could not only increase household income but also stimulate the local rural economy. Given the positive coefficients for the basic needs index and the advanced needs index, it can be induced that cash crop cultivation could significantly enhance the quality of life for the rural households. The four cash crop types presented two opposite correlations with the emigration index. The opposite coefficients suggested that tea and mulberry cultivation would accelerate the emigration process while fruit and nursery would slow

down the emigration rate in the rural areas.

## 4. Discussion and conclusions

### 4.1. Socioeconomic impacts of cash crop cultivation

Our study indicated that the economic benefits of cash crop cultivation were evident, including raising household income, boosting rural economy, increasing fiscal revenues, and attracting foreign investment. These results were consistent with prior investigations in tropical places (Bunyamin, 2008; Castiblanco et al., 2015; Novo et al., 2010; Sheil et al., 2009) and supported the affirmation that cash cropping can alleviate rural poverty (GLOBAL, 2011; Rist et al., 2010). The household income in the sugar cane cultivated regions was at least 30% higher than that in other rural areas of Brazil (Novo et al., 2010). It was found that oil palm cultivations contributed to increases in household incomes and provincial gross domestic product in Sumatra and Kalimantan (Bunyamin, 2008). In Indonesia, farmers involved in oil palm cultivations earned 7 times higher than subsistence farmers (Sheil et al., 2009). Castiblanco et al. (2015) discovered that oil palm expansion led to an increase in municipal fiscal income. In the subtropical China, cash crops are more profitable than the staple grains. For example, the economic profit per hectare of tea and fruit is more than three times higher than that of paddy fields. Therefore, the market advantage of cash crops is the essential to the realization of economic gains associated with cash crop cultivation. Besides, cash crop product processing has a longer industry chain compared to the traditional staple grains. In Hangzhou region, modern industrial systems have been formed to take deep processing of cash crops (e.g., tea, mulberry and fruit). It has attracted large volume of foreign investment in the downstream food and tourism industry.

From the social perspectives, potential positive benefits yielded by cash crop cultivation to the rural communities include infrastructure improvements, job creations and land value increases (Dufey, 2008; German et al., 2011; World Bank, 2010). For example, the oil palm industry on average generates 0.4 jobs per ha in Malaysia and Indonesia, which is much higher than that of the traditional smallholder production (0.05 jobs per ha) (World Bank, 2010). Our findings support the previous statement, as cash crop expansion was evidence to have positive relationships between social welfare and life quality indicators. Economic gaining plays a mediating role in explaining the social impacts of cash crop cultivation, as income is a necessity to enhance life quality. The local government, driven by economic incentives, called on

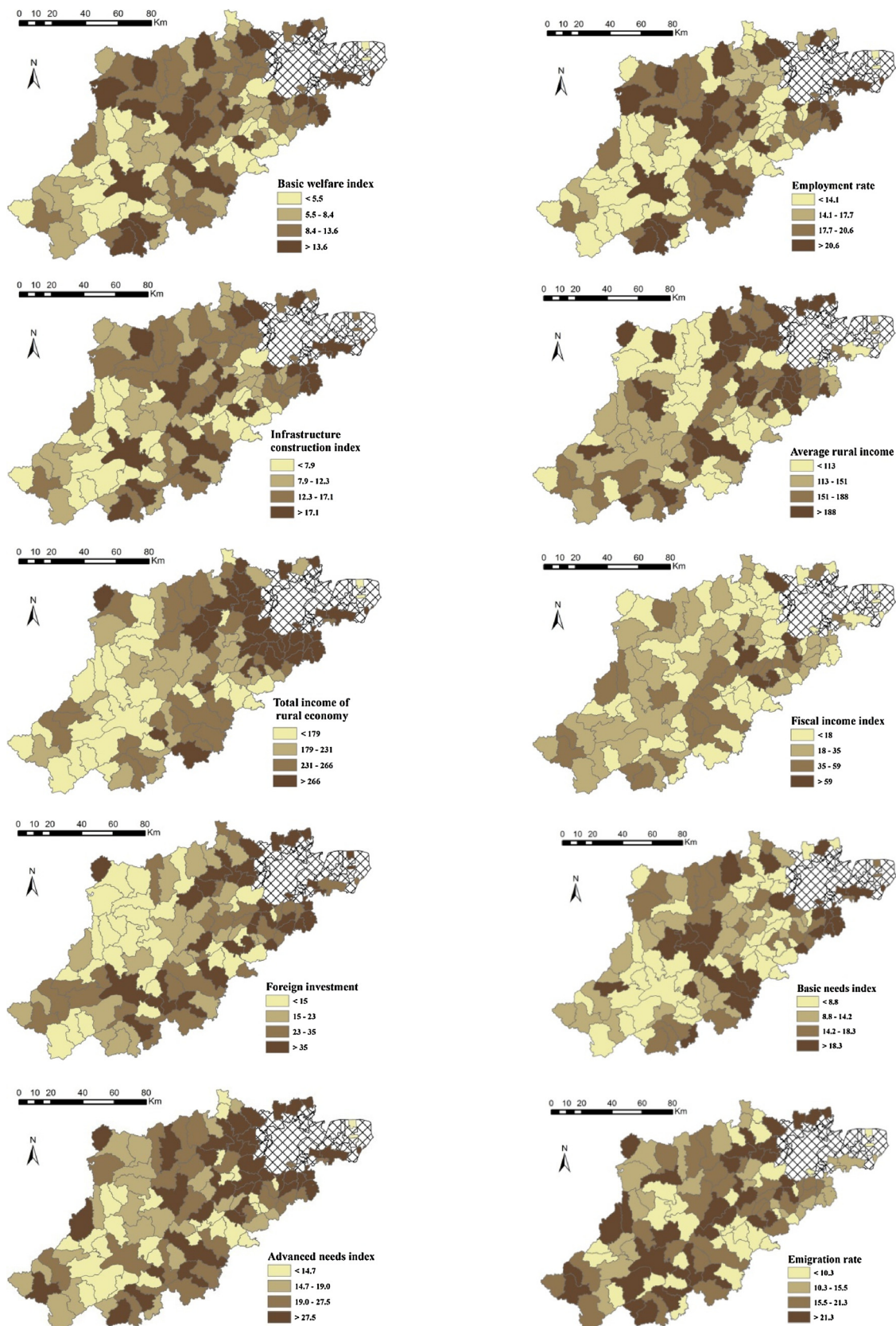


Fig. 7. Changes of socioeconomic indicators at town level from 2004 and 2014 within Hangzhou region, China (unit: %).



**Table 3**  
Relationships between changes of landscape-level metrics and cash crop expansion identified by spatial regression<sup>a</sup>.

| Y       | X        | Coefficients    | Spatial dependency                  | R <sup>2</sup>     |
|---------|----------|-----------------|-------------------------------------|--------------------|
| PD      | Tea      | 0.411           | W <sub>Y</sub> <sup>b</sup> (1.423) | 0.42 <sup>**</sup> |
|         | Fruit    | 0.291           | W <sub>Y</sub> <sup>b</sup> (1.856) | 0.38 <sup>**</sup> |
|         | Mulberry | 0.204           | W <sub>Y</sub> <sup>b</sup> (1.047) | 0.24 <sup>**</sup> |
|         | Nursery  | 0.309           | W <sub>Y</sub> <sup>b</sup> (2.356) | 0.31 <sup>**</sup> |
| LSI     | Tea      | 0.369           | W <sub>Y</sub> <sup>b</sup> (0.884) | 0.36 <sup>**</sup> |
|         | Fruit    | 0.315           | W <sub>Y</sub> <sup>b</sup> (1.025) | 0.39 <sup>**</sup> |
|         | Mulberry | 0.232           | W <sub>Y</sub> <sup>b</sup> (0.997) | 0.19 <sup>**</sup> |
|         | Nursery  | 0.442           | W <sub>Y</sub> <sup>b</sup> (1.224) | 0.32 <sup>**</sup> |
| CONNECT | Tea      | −0.143          | W <sub>Y</sub> <sup>b</sup> (2.001) | 0.29 <sup>**</sup> |
|         | Fruit    | −0.093          | W <sub>Y</sub> <sup>b</sup> (1.849) | 0.22 <sup>**</sup> |
|         | Mulberry | NS <sup>c</sup> |                                     |                    |
|         | Nursery  | −0.125          | W <sub>Y</sub> <sup>b</sup> (2.446) | 0.26 <sup>**</sup> |
| SPLIT   | Tea      | 0.164           | W <sub>Y</sub> <sup>b</sup> (1.324) | 0.39 <sup>**</sup> |
|         | Fruit    | 0.422           | W <sub>Y</sub> <sup>b</sup> (2.678) | 0.47 <sup>**</sup> |
|         | Mulberry | 0.101           | W <sub>Y</sub> <sup>b</sup> (1.541) | 0.21 <sup>**</sup> |
|         | Nursery  | 0.116           | W <sub>Y</sub> <sup>b</sup> (1.333) | 0.23 <sup>**</sup> |
| SHDI    | Tea      | 0.028           | Lambda <sup>d</sup> = 1.441         | 0.25 <sup>**</sup> |
|         | Fruit    | 0.058           | Lambda <sup>d</sup> = 1.676         | 0.19 <sup>**</sup> |
|         | Mulberry | NS <sup>c</sup> |                                     |                    |
|         | Nursery  | 0.178           | Lambda <sup>d</sup> = 1.243         | 0.20 <sup>**</sup> |

Abbreviations: patch density (PD); landscape shape index (LSI); connectance index (CONNECT); split index (SPLIT); Shannon's diversity index (SHDI).

<sup>\*\*</sup>  $p < 0.01$ .

<sup>a</sup> N = 106 for tea; N = 121 for fruit; N = 114 for mulberry; N = 117 for nursery.

<sup>b</sup> Spatial lag regression: W<sub>Y</sub> denotes the spatial matrix for dependent variable.

<sup>c</sup> NS: no significant relationship.

<sup>d</sup> Spatial error regression.

**Table 5**  
Relationships between changes of ecosystem service values and cash crop expansion identified by spatial regression<sup>a</sup>.

| Y                             | X        | Coefficient | Spatial dependency                  | R <sup>2</sup>     |
|-------------------------------|----------|-------------|-------------------------------------|--------------------|
| Gas regulation                | Tea      | −0.126      | W <sub>Y</sub> <sup>b</sup> (0.441) | 0.21 <sup>**</sup> |
|                               | Fruit    | −0.155      | W <sub>Y</sub> <sup>b</sup> (0.502) | 0.24 <sup>**</sup> |
|                               | Mulberry | −0.072      | W <sub>Y</sub> <sup>b</sup> (0.367) | 0.17 <sup>**</sup> |
|                               | Nursery  | −0.034      | W <sub>Y</sub> <sup>b</sup> (0.380) | 0.08 <sup>**</sup> |
| Climate regulation            | Tea      | −0.113      | W <sub>Y</sub> <sup>b</sup> (0.216) | 0.23 <sup>**</sup> |
|                               | Fruit    | −0.208      | W <sub>Y</sub> <sup>b</sup> (0.309) | 0.25 <sup>**</sup> |
|                               | Mulberry | −0.022      | W <sub>Y</sub> <sup>b</sup> (0.354) | 0.07 <sup>**</sup> |
|                               | Nursery  | −0.013      | W <sub>Y</sub> <sup>b</sup> (0.296) | 0.05 <sup>**</sup> |
| Water reservation             | Tea      | −0.105      | W <sub>Y</sub> <sup>b</sup> (0.084) | 0.18 <sup>**</sup> |
|                               | Fruit    | −0.157      | W <sub>Y</sub> <sup>b</sup> (0.123) | 0.13 <sup>**</sup> |
|                               | Mulberry | −0.087      | W <sub>Y</sub> <sup>b</sup> (0.169) | 0.08 <sup>**</sup> |
|                               | Nursery  | −0.069      | W <sub>Y</sub> <sup>b</sup> (0.153) | 0.06 <sup>**</sup> |
| Soil formation and protection | Tea      | −0.128      | W <sub>Y</sub> <sup>b</sup> (0.182) | 0.19 <sup>**</sup> |
|                               | Fruit    | −0.087      | W <sub>Y</sub> <sup>b</sup> (0.357) | 0.15 <sup>**</sup> |
|                               | Mulberry | −0.033      | W <sub>Y</sub> <sup>b</sup> (0.367) | 0.09 <sup>**</sup> |
|                               | Nursery  | −0.024      | W <sub>Y</sub> <sup>b</sup> (0.284) | 0.07 <sup>**</sup> |
| Biodiversity conservation     | Tea      | −0.022      | W <sub>Y</sub> <sup>b</sup> (0.238) | 0.23 <sup>**</sup> |
|                               | Fruit    | −0.034      | W <sub>Y</sub> <sup>b</sup> (0.349) | 0.21 <sup>**</sup> |
|                               | Mulberry | −0.017      | W <sub>Y</sub> <sup>b</sup> (0.405) | 0.11 <sup>**</sup> |
|                               | Nursery  | −0.012      | W <sub>Y</sub> <sup>b</sup> (0.321) | 0.07 <sup>**</sup> |

<sup>\*\*</sup>  $p < 0.01$ .

<sup>a</sup> N = 106 for tea; N = 121 for fruit; N = 114 for mulberry; N = 117 for nursery.

<sup>b</sup> Spatial lag regression: W<sub>Y</sub> denotes the spatial matrix for dependent variable.

the involvement of households in cash crop cultivation and promoted the cash crop market. Consequently, great efforts were spared by the local government to improve the living conditions and infrastructure, in order to accelerate circulation and absorb foreign capital. Our study

**Table 4**  
Relationships between changes of class-level metrics and cash crop expansion identified by spatial regression<sup>a</sup>.

| Y (forest) | X        | Coefficient     | Spatial dependency                  | R <sup>2</sup>     | Y (farmland) | X        | Coefficient     | Spatial dependency                  | R <sup>2</sup>     |
|------------|----------|-----------------|-------------------------------------|--------------------|--------------|----------|-----------------|-------------------------------------|--------------------|
| TA         | Tea      | −0.173          | W <sub>Y</sub> <sup>b</sup> (0.424) | 0.53 <sup>**</sup> | TA           | Tea      | −0.072          | W <sub>Y</sub> <sup>b</sup> (2.431) | 0.07 <sup>**</sup> |
|            | Fruit    | −0.225          | W <sub>Y</sub> <sup>b</sup> (0.589) | 0.44 <sup>**</sup> |              | Fruit    | −0.162          | W <sub>Y</sub> <sup>b</sup> (1.387) | 0.18 <sup>**</sup> |
|            | Mulberry | −0.034          | W <sub>Y</sub> <sup>b</sup> (0.353) | 0.12 <sup>**</sup> |              | Mulberry | −0.064          | W <sub>Y</sub> <sup>b</sup> (1.861) | 0.13 <sup>**</sup> |
|            | Nursery  | −0.157          | W <sub>Y</sub> <sup>b</sup> (0.457) | 0.24 <sup>**</sup> |              | Nursery  | −0.189          | W <sub>Y</sub> <sup>b</sup> (1.747) | 0.24 <sup>**</sup> |
| ED         | Tea      | 0.306           | W <sub>Y</sub> <sup>b</sup> (0.754) | 0.36 <sup>**</sup> | ED           | Tea      | 0.162           | W <sub>Y</sub> <sup>b</sup> (0.967) | 0.06 <sup>**</sup> |
|            | Fruit    | 0.186           | W <sub>Y</sub> <sup>b</sup> (1.057) | 0.28 <sup>**</sup> |              | Fruit    | 0.072           | W <sub>Y</sub> <sup>b</sup> (1.055) | 0.19 <sup>**</sup> |
|            | Mulberry | 0.085           | W <sub>Y</sub> <sup>b</sup> (0.497) | 0.13 <sup>**</sup> |              | Mulberry | 0.159           | W <sub>Y</sub> <sup>b</sup> (1.227) | 0.15 <sup>**</sup> |
|            | Nursery  | 0.154           | W <sub>Y</sub> <sup>b</sup> (0.964) | 0.23 <sup>**</sup> |              | Nursery  | 0.214           | W <sub>Y</sub> <sup>b</sup> (0.850) | 0.23 <sup>**</sup> |
| ENND       | Tea      | −0.056          | Lambda <sup>d</sup> = 2.441         | 0.25 <sup>**</sup> | ENND         | Tea      | NS <sup>c</sup> |                                     |                    |
|            | Fruit    | −0.026          | Lambda <sup>d</sup> = 1.579         | 0.16 <sup>**</sup> |              | Fruit    | −0.134          | Lambda <sup>d</sup> = 1.257         | 0.16 <sup>**</sup> |
|            | Mulberry | NS <sup>c</sup> |                                     |                    |              | Mulberry | NS <sup>c</sup> |                                     |                    |
|            | Nursery  | −0.033          | Lambda <sup>d</sup> = 1.852         | 0.19 <sup>**</sup> |              | Nursery  | −0.155          | Lambda <sup>d</sup> = 1.367         | 0.19 <sup>**</sup> |
| AWSI       | Tea      | 0.166           | W <sub>Y</sub> <sup>b</sup> (1.231) | 0.40 <sup>**</sup> | AWSI         | Tea      | 0.074           | W <sub>Y</sub> <sup>b</sup> (0.682) | 0.11 <sup>**</sup> |
|            | Fruit    | 0.029           | W <sub>Y</sub> <sup>b</sup> (1.553) | 0.31 <sup>**</sup> |              | Fruit    | 0.159           | W <sub>Y</sub> <sup>b</sup> (0.843) | 0.22 <sup>**</sup> |
|            | Mulberry | 0.017           | W <sub>Y</sub> <sup>b</sup> (0.974) | 0.14 <sup>**</sup> |              | Mulberry | 0.088           | W <sub>Y</sub> <sup>b</sup> (0.941) | 0.16 <sup>**</sup> |
|            | Nursery  | 0.196           | W <sub>Y</sub> <sup>b</sup> (1.718) | 0.25 <sup>**</sup> |              | Nursery  | 0.171           | W <sub>Y</sub> <sup>b</sup> (1.247) | 0.27 <sup>**</sup> |
| AI         | Tea      | −0.165          | W <sub>Y</sub> <sup>b</sup> (1.334) | 0.32 <sup>**</sup> | AI           | Tea      | NS <sup>c</sup> |                                     |                    |
|            | Fruit    | −0.119          | W <sub>Y</sub> <sup>b</sup> (1.418) | 0.29 <sup>**</sup> |              | Fruit    | −0.127          | W <sub>Y</sub> <sup>b</sup> (1.357) | 0.18 <sup>**</sup> |
|            | Mulberry | −0.075          | W <sub>Y</sub> <sup>b</sup> (1.563) | 0.15 <sup>**</sup> |              | Mulberry | −0.087          | W <sub>Y</sub> <sup>b</sup> (1.189) | 0.14 <sup>**</sup> |
|            | Nursery  | −0.108          | W <sub>Y</sub> <sup>b</sup> (0.746) | 0.26 <sup>**</sup> |              | Nursery  | −0.187          | W <sub>Y</sub> <sup>b</sup> (1.057) | 0.21 <sup>**</sup> |

Abbreviations: total area (TA), edge density (ED), mean Euclidean nearest neighbour distance (ENND), area weighted shape index (AWSI), and aggregation index (AI).

<sup>\*\*</sup>  $p < 0.01$ .

<sup>a</sup> N = 106 for tea; N = 121 for fruit; N = 114 for mulberry; N = 117 for nursery.

<sup>b</sup> Spatial lag regression: W<sub>Y</sub> denotes the spatial matrix for dependent variable.

<sup>c</sup> NS: no significant relationship.

<sup>d</sup> Spatial error regression.

**Table 6**  
Relationships between changes of socioeconomic indicators and cash crop expansion identified by spatial regression<sup>a</sup>.

| Y                                 | X        | Coefficient     | Spatial dependency                  | R <sup>2</sup> |
|-----------------------------------|----------|-----------------|-------------------------------------|----------------|
| Basic welfare index               | Tea      | 0.195           | W <sub>Y</sub> <sup>b</sup> (1.244) | 0.16**         |
|                                   | Fruit    | 0.206           | W <sub>Y</sub> <sup>b</sup> (1.032) | 0.19**         |
|                                   | Mulberry | 0.087           | W <sub>Y</sub> <sup>b</sup> (1.019) | 0.11**         |
|                                   | Nursery  | 0.115           | W <sub>Y</sub> <sup>b</sup> (1.153) | 0.08**         |
| Employment rate                   | Tea      | 0.106           | W <sub>Y</sub> <sup>b</sup> (0.852) | 0.23**         |
|                                   | Fruit    | 0.186           | W <sub>Y</sub> <sup>b</sup> (0.961) | 0.21**         |
|                                   | Mulberry | 0.087           | W <sub>Y</sub> <sup>b</sup> (0.743) | 0.10**         |
|                                   | Nursery  | NS <sup>c</sup> |                                     |                |
| Infrastructure construction index | Tea      | 0.157           | Lambda <sup>d</sup> = 1.524         | 0.26**         |
|                                   | Fruit    | 0.128           | Lambda <sup>d</sup> = 1.087         | 0.21**         |
|                                   | Mulberry | 0.103           | Lambda <sup>d</sup> = 1.917         | 0.15**         |
|                                   | Nursery  | 0.119           | Lambda <sup>d</sup> = 1.262         | 0.23**         |
| Average rural income              | Tea      | 0.369           | W <sub>Y</sub> <sup>b</sup> (1.457) | 0.49**         |
|                                   | Fruit    | 0.352           | W <sub>Y</sub> <sup>b</sup> (1.068) | 0.55**         |
|                                   | Mulberry | 0.281           | W <sub>Y</sub> <sup>b</sup> (1.226) | 0.39**         |
|                                   | Nursery  | 0.266           | W <sub>Y</sub> <sup>b</sup> (1.187) | 0.41**         |
| Total income of rural economy     | Tea      | 0.207           | W <sub>Y</sub> <sup>b</sup> (2.357) | 0.21**         |
|                                   | Fruit    | 0.175           | W <sub>Y</sub> <sup>b</sup> (2.069) | 0.27**         |
|                                   | Mulberry | 0.084           | W <sub>Y</sub> <sup>b</sup> (1.852) | 0.15**         |
|                                   | Nursery  | 0.111           | W <sub>Y</sub> <sup>b</sup> (1.927) | 0.20**         |
| Fiscal income index               | Tea      | 0.094           | W <sub>Y</sub> <sup>b</sup> (0.885) | 0.12**         |
|                                   | Fruit    | 0.135           | W <sub>Y</sub> <sup>b</sup> (0.751) | 0.14**         |
|                                   | Mulberry | 0.057           | W <sub>Y</sub> <sup>b</sup> (0.526) | 0.09**         |
|                                   | Nursery  | 0.066           | W <sub>Y</sub> <sup>b</sup> (0.509) | 0.07**         |
| Foreign investment                | Tea      | 0.129           | Lambda <sup>d</sup> = 0.654         | 0.23**         |
|                                   | Fruit    | 0.088           | Lambda <sup>d</sup> = 0.863         | 0.10**         |
|                                   | Mulberry | 0.043           | Lambda <sup>d</sup> = 0.710         | 0.22**         |
|                                   | Nursery  | 0.057           | Lambda <sup>d</sup> = 1.29          | 0.07**         |
| Basic needs index                 | Tea      | 0.103           | W <sub>Y</sub> <sup>b</sup> (0.214) | 0.14**         |
|                                   | Fruit    | 0.157           | W <sub>Y</sub> <sup>b</sup> (0.231) | 0.18**         |
|                                   | Mulberry | 0.122           | W <sub>Y</sub> <sup>b</sup> (0.195) | 0.15**         |
|                                   | Nursery  | NS <sup>c</sup> | W <sub>Y</sub> <sup>b</sup> (0.175) |                |
| Advanced needs index              | Tea      | 0.203           | W <sub>Y</sub> <sup>b</sup> (1.025) | 0.31**         |
|                                   | Fruit    | 0.153           | W <sub>Y</sub> <sup>b</sup> (1.103) | 0.27**         |
|                                   | Mulberry | 0.128           | W <sub>Y</sub> <sup>b</sup> (1.379) | 0.16**         |
|                                   | Nursery  | 0.105           | W <sub>Y</sub> <sup>b</sup> (0.985) | 0.22**         |
| Emigration rate                   | Tea      | 0.149           | Lambda <sup>d</sup> = 1.597         | 0.23**         |
|                                   | Fruit    | −0.204          | Lambda <sup>d</sup> = 1.005         | 0.13**         |
|                                   | Mulberry | 0.231           | Lambda <sup>d</sup> = 2.235         | 0.24**         |
|                                   | Nursery  | −0.146          | Lambda <sup>d</sup> = 1.621         | 0.15**         |

\*\*  $p < 0.01$ .

<sup>a</sup> N = 106 for tea; N = 121 for fruit; N = 114 for mulberry; N = 117 for nursery.

<sup>b</sup> Spatial lag regression: W<sub>Y</sub> denotes the spatial matrix for dependent variable.

<sup>c</sup> NS: no significant relationship.

<sup>d</sup> Spatial error regression.

also discovered that the social impacts varied with cash crop types. Tea, fruit and mulberry cultivation was found to increase employment, while nursery plantation had no significant influence. It makes sense in that the business chain of nursery cultivation is short. In most cases, the seedlings and plants are directly transported from the supplier to the demanders. In addition, tea and mulberry cultivations would accelerate the emigration process while fruit and nursery would slow down the emigration rate in the rural areas. Such discrepancies should be related to the characteristics of rural labors involved in the planting of the four cash crop types. Tea and mulberry cultivation consumes relatively less labor and their growing seasons are shorter. Women and the elder usually act as the major labor for tea and mulberry planting, and the young male laborers choose to migrate into urban cities in seek of more earnings.

## 4.2. Ecological consequences of cash crop cultivations

Regarding the ecological consequences of cash crop cultivation in tropical regions, many scholars have claimed that the most apparent aspect was the replacement of forest in the non-marginal areas (Castiblanco et al., 2013; Miyamoto, 2006; Zhai et al., 2012; Ziegler et al., 2009). Our analysis demonstrated that forest and farmland were both victims of cash crop cultivation in subtropical China. In particular, the expanded tea and fruit plantations generally occurred at the forested land at steep slope. Such land conversion was evidenced to be a critical contributor to soil erosion in the study area (Liu et al., 2012). Consequently, it was expected that soil erosion would be accelerated in the cash crop cultivated areas within Hangzhou region. The mulberry and nursery expanded at the expense of farmland with high soil quality in the study area. It would promote the competition around supplies of agricultural production and threat food security. This result supported the argument that cash crop monoculture would raise the issue of food security (Yi et al., 2014b). Establishing new cash crop plantations, in general terms, was the spontaneous behavior of the local farmers (Zhang et al., 2014). Cash crop expansion lacked long-term guide and strategical planning (Su et al., 2014a), failing to form an orderly land use pattern that could alleviate the pressures on ecosystems. It further led to the irregularity, isolation and instability of forest and farmland patches. Forest disturbances would lead to reductions in species diversity and in the prevalence of high conservation value species (Fitzherbert et al., 2008). The biodiversity in the study area should be threatened as a consequence of cash crop cultivation. Besides, the isolated and irregular farmland is considered to be unsuitable for utilization. The isolated patches are probably to be abandoned (Su et al., 2014a), further increasing the vulnerability to food security. All these analysis implied the negative ecological costs associated with the shift to cash crop monoculture.

At the landscape level, cash crop expansion would significantly lead to landscape fragmentation, isolation and irregularity. These results accorded with the previous findings that cash crop cultivation would fragment traditional rural landscapes (Godone et al., 2014). Landscape fragmentation has been evidenced to substantially affect various ecological processes (Girvetz et al., 2008; Godone et al., 2014; Uuemaa et al., 2013) and the provision of ecosystem services (Su et al., 2012). Our study highlighted that ESV would significantly decline associated with cash cultivation. The negative coefficients for water conservation service and biodiversity conservation service supported the deduction that cash crop cultivation would accelerate soil erosion and reduce biodiversity (Cha, 2005; German et al., 2011; Yi et al., 2014a,b). Past literature demonstrated that cash crop expansion could lead to reduced carbon stocks and increased greenhouse gas (Petsri et al., 2013). This paper provided additional insights into this issue, as cash crop expansion was found to have negative relationships with the gas regulation service and climate regulation service. All these results implied that cash crop cultivation could significantly impair ecosystem functions to deliver services.

## 4.3. Recommendations on land use policy

Hangzhou is one of the major cash crop planation bases in China. Cash crop cultivation was primarily affected by two converse requirements. With an attempt to promote forest conservation, the central government has enacted a series of policies, including the Grain to Green Program, the Ecological City Program, and the Natural Forest Conversation Program. However, owing to the raising household income of cultivating cash crop, the profit-maximizing smallholder farmers are unwilling to adopt the cash crop restriction practices without external encouragement. Under China's land tenure policy, households are granted rights to independently decide land use types. It results in economic benefit at the cost of ecological security. In this context, the government and households are confronted with conflicts

over enlarging cash crop cultivation. It is not a plausible solution that merely prevent households from enlarging cash crop cultivation to achieve forests conservation. Aiming at achieving a balance between economic gain and ecological conservation, it is crucial and critical to enact appropriate land use policies.

In view of the dilemma mentioned above, we propose three potential options towards the balance between economic benefit and ecological conservation regarding cash crop cultivation. Initially, education programs can be launched to impart knowledge and skills of the best management practices. Not only can farmers recognize the importance of ecological conservation, but also they can obtain knowledge base associated with how to manage cash crop gardens for fulfilling sustainability criteria, including soil erosion prevention, fertilizer and herbicide control. In addition, never should we ignore that rational cash expansion legitimization can exclude primary forests from excessive deforestation. Particularly, a dual appraisal system can be introduced to evaluate the comparative importance of economic benefit and ecological cost. Last but not least, financial subsidies can be adopted for forest restoration to compensate for income loss with respect to cash crop restriction, and tax revenue for cash crop consumption and specific national funds are potential financial resources.

#### 4.4. Methodological discussion

This paper quantified the socioeconomic and ecological consequences of cash crop cultivation by employing multiple approaches, including remote sensing, landscape metric analysis, ESV evaluation and spatial regression. In particular, a town level statistical dataset was accessed and several indicators were formulated to facilitate the quantification of socioeconomic impacts of cash crop cultivation.

#### Appendix A. Ecological significance of landscape metrics

|                 | Metrics   | Ecological significance                                                                                                                                                                                                                 |
|-----------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Landscape level | PD, SPLIT | Indicators of fragmentation. Fragmentation is destructive to the substance circulation, information transfer and energy exchanges within and between ecosystems.                                                                        |
|                 | LSI       | Indicator of shape complexity. Complex shaped landscape patterns denote higher disturbances in ecosystem functions.                                                                                                                     |
|                 | CONNECT   | Indicator of connectivity. Higher connectivity denotes higher proximity between patches of the same class. It benefits migratory and reproductive patterns of the wildlife.                                                             |
|                 | SHDI      | Indicator of simplification or heterogeneity. For highly disturbed landscapes, higher SHDI denotes the richness of different land use types and fragmentation. For conserved landscapes, higher SHDI always denote higher biodiversity. |
| Class level     | TA        | Indicator of dominance. It restricts the abundance, quantity and food chain of species which rely on certain land type as habitat.                                                                                                      |
|                 | ED        | Indicator of stability. Higher ED denotes edge effect associated with heterogeneous landscape patters. Patches with high ED are instable and vulnerable to disturbance.                                                                 |
|                 | ENND      | Indicator of distance between certain class patches. Shorter ENND denotes higher probability of interference among patches and spreading of disturbance.                                                                                |
|                 | AWSI      | Indicator of shape irregularity. Complex shaped patches restrict the migration and foraging activities of animals and affect crop cultivation and production efficiency.                                                                |
|                 | AI        | Indicator of aggregation or spreading. Higher AI denotes more dominance and connectivity.                                                                                                                                               |

Abbreviations: patch density (PD), landscape shape index (LSI), connectance index (CONNECT), split index (SPLIT) and Shannon's diversity index (SHDI), total area (TA), edge density (ED), mean Euclidean nearest neighbour distance (ENND), area weighted shape index (AWSI), aggregation index (AI).

#### References

- Anselin, L., 1988. *Spatial Econometrics: Methods and Models*. Kluwer Academic Publishers, Dordrecht, The Netherlands.
- Anselin, L., Syabri, I., Kho, Y., 2006. GeoDa: an introduction to spatial data analysis. *Geogr. Anal.* 38, 5–22.
- Barkmann, J., Burkard, G., Faust, H., Fremerey, M., Koch, S., Lanini, A., 2010. Landtenure rights, village institutions, and rainforest conversion in Central Sulawesi (Indonesia). In: Tschamtk, T., Leuschner, C., Veldkamp, E., Faust, H., Guhardja, E., Bidin, A. (Eds.), *Tropical Rainforests and Agroforests Under Global Change*. Springer, Berlin, pp. 141–160.
- Bunyamin, B., 2008. *Impacts of Oil Palm Plantations on the Regional Economy of West Kalimantan Region*. Untan Press, Pontianak, Indonesia.
- Carlson, K.M., Curran, L.M., Ratnasari, D., Pittman, A.M., Soares-Filho, B.S., Asner, G.P., 2012. Committed carbon emissions, deforestation, and community land conversion from oil palm plantation expansion in West Kalimantan, Indonesia. *Proc. Natl. Acad. Sci.* 109, 7559–7564.
- Castiblanco, C., Etter, A., Aide, T.M., 2013. Oil palm plantations in Colombia: a model of



- future expansion. *Environ. Sci. Policy* 27, 172–183.
- Castiblanco, C., Etter, A., Ramirez, A., 2015. Impacts of oil palm expansion in Colombia: what do socioeconomic indicators show? *Land Use Policy* 44, 31–43.
- Cha, Z., 2005. Sustainable land management practices for rubber plantations in mountainous area of Hainan. *Pedosphere* 15, 404–408.
- Costanza, R., d'Arge, R., de Groot, R., Farber, S., Grasso, M., Hannon, B., et al., 1997. The value of the world's ecosystem services and natural capital. *Nature* 387, 253–260.
- Dal Belo Leite, J.G., Justino, F.B., Silva, J.V., Florin, M.J., van Itersum, M.K., 2015. Socioeconomic and environmental assessment of biodiesel crops on family farming systems in Brazil. *Agric. Syst.* 133, 22–34.
- Delpuech, C., Lebouis, A., 2014. The elusive quest for supply response to cash-crop market reforms in Sub-Saharan Africa: the case of cotton. *World Dev.* 64, 521–537.
- Dong, J., Xiao, X., Chen, B., Torbick, N., Jin, C., Zhang, G., Biradar, C., 2013. Mapping deciduous rubber plantations through integration of PALSAR and multi-temporal Landsat imagery. *Remote Sens. Environ.* 134, 392–402.
- Dufey, A., 2008. Impacts of sugarcane bioethanol towards the millennium development goals. Pages 199–225 In: Zurbier, P., van de Vooren, J. (Eds.), *Sugarcane Ethanol: Contributions to Climate Change Mitigation and the Environment*. Wageningen Academic Publishers, Wageningen, The Netherlands.
- Ewing, M., Msangi, S., 2009. Biofuels production in developing countries: assessing tradeoffs in welfare and food security. *Environ. Sci. Policy* 12, 520–528.
- Fernandes, M.R., Aguiar, F.C., Ferreira, M.T., 2011. Assessing riparian vegetation structure and the influence of land use using landscape metrics and geostatistical tools. *Land Urban Plan.* 99, 166–177.
- Fitzherbert, E.B., Strubbig, M.J., Morel, A., Danielsen, F., Brühl, C.A., Donald, P.F., Phalan, B., 2008. How will oil palm expansion affect biodiversity? *Trends Ecol. Evol.* 23, 538–545.
- Gatto, M., Wollni, M., Qaim, M., 2015. Oil palm boom and land-use dynamics in Indonesia: the role of policies and socioeconomic factors. *Land Use Policy* 46, 292–303.
- German, L., Schoneveld, G., Pacheco, P., 2011. The social and environmental impacts of biofuel feedstock cultivation: evidence from multi-site research in the forest. *Front. Ecol. Soc.* 16, 3–24.
- GLOBAL, ITS, 2011. *The Economic Benefits of Palm Oil in Papua New Guinea*. International Trade Strategies, Melbourne.
- Girvetz, E.H., Thorne, J.H., Berry, A.M., Jaeger, J.A.G., 2008. Integration of landscape fragmentation analysis into regional planning: a statewide multiscale case study for California, USA. *Landscape Urban Plan.* 86, 205–218.
- Godone, D., Garbarino, M., Sibona, E., Garnerio, G., Godone, F., 2014. Progressive fragmentation of a traditional Mediterranean landscape by hazelnut plantations: the impact of CAP over time in the Langhe region (NW Italy). *Land Use Policy* 36, 259–266.
- Kromroy, K., Ward, K., Castillo, P., Juzwik, J., 2007. Relationships between urbanization and the oak resource of the Minneapolis/St. Paul metropolitan area from 1991 to 1998. *Landscape Urban Plan.* 80, 375–385.
- Solon, J., 2009. Spatial context of urbanization: landscape pattern and changes between 1950 and 1990 in the Warsaw metropolitan area, Poland. *Landscape Urban Plan.* 93, 250–261.
- LeSage, J., Pace, R.K., 2009. *Introduction to Spatial Econometrics*. Taylor & Francis/CRC, London.
- Leitão, A.B., Ahern, J., 2002. Applying landscape ecological concepts and metrics in sustainable landscape planning. *Landscape Urban Plan.* 59, 65–93.
- Li, Z., Fox, J.M., 2012. Mapping rubber tree growth in mainland Southeast Asia using time-series MODIS 250 m NDVI and statistical data. *Appl. Geogr.* 32, 420–432.
- Liu, Z., Yang, J., Yang, Z., Zou, J., 2012. Effects of rainfall and fertilizer types on nitrogen and phosphorus concentrations in surface runoff from subtropical tea fields in Zhejiang, China. *Nutr. Cycl. Agroecosyst.* 93, 297–307.
- Mairota, P., Cafarelli, B., Boccaccio, L., Leroni, V., Labadessa, R., Kosmidou, V., Nagendra, H., 2013. Using landscape structure to develop quantitative baselines for protected area monitoring. *Ecol. Indic.* 33, 82–95.
- McCarthy, J., 2010. Processes of inclusion and adverse incorporation: oil palm and agrarian change in Sumatra, Indonesia. *J. Peasant Stud.* 37, 821–850.
- Millennium Ecosystem Assessment, 2003. *Ecosystems and Human Well-Being: Current State and Trends*. Island Press, Washington, DC.
- Miyamoto, M., 2006. Forest conversion to rubber around Sumatran villages in Indonesia: comparing the impacts of road construction, transmigration projects and population. *For. Policy Econ.* 9, 1–12.
- Novo, A., Jansen, K., Slingerland, M., Giller, K., 2010. Biofuel, dairy production and beef in Brazil: competing claims on land use in São Paulo state. *J. Peasant Stud.* 37, 769–792.
- Obidzinski, K., Andriani, R., Komarudin, H., Andrianto, A., 2012. Environmental and social impacts of oil palm plantations and their implications for biofuel production in Indonesia. *Ecol. Soc.* 17, 1–25.
- Petsri, S., Chidthaisong, A., Pumijumnong, N., Wachrinrat, C., 2013. Greenhouse gas emissions and carbon stock changes in rubber tree plantations in Thailand from 1990 to 2004. *J. Clean. Prod.* 52, 61–70.
- Pôças, I., Cunha, M., Pereira, L.S., 2011. Remote sensing based indicators of changes in a mountain rural landscape of Northeast Portugal. *Appl. Geogr.* 31, 871–880.
- Qiu, J., 2009. Where the rubber meets the garden. *Nature* 457, 246–247.
- Rempel, R., 2008. *Patch Analyst 4. Computer Software Program Produced by the author at Centre for Northern Forest Ecosystem Research*. Lakehead University Campus. <http://flash.lakeheadu.ca/wrrempe/patch/download.htm>.
- Ribeiro, S.C., Lovett, A., 2009. Associations between forest characteristics and socioeconomic development: a case study from Portugal. *J. Environ. Manage.* 90, 2873–2881.
- Rist, L., Feintrenie, L., Levang, P., 2010. The livelihood impacts of oil palm: smallholders in Indonesia. *Biodivers. Conserv.* 19, 1009–1024.
- Sheil, D., Casson, A., Meijaard, E., van Noordwijk, M., Gaskell, J., Sunderland-Groves, J., Wertz, K., Kanninen, M., 2009. The Impacts and Opportunities of Oil Palm in Southeast Asia What Do We Know and What Do We Need to Know? CIFOR, Indonesia.
- Sowińska-Świerkosz, B.N., Soszyński, D., 2014. Landscape structure versus the effectiveness of nature conservation: Roztocze region case study (Poland). *Ecol. Indic.* 43, 143–153.
- Su, S., Jiang, Z., Zhang, Q., Zhang, Y., 2011. Transformation of agricultural landscapes under rapid urbanization: a threat to sustainability in Hang-Jia-Hu region, China. *Appl. Geogr.* 31, 439–449.
- Su, S., Wang, Y., Luo, F., Mai, G., Pu, J., 2014a. Peri-urban vegetated landscape pattern changes in relation to socioeconomic development. *Ecol. Indic.* 46, 477–486.
- Su, S., Li, D., Hu, Y., Xiao, R., Zhang, Y., 2014b. Spatially non-stationary response of ecosystem service value changes to urbanization in Shanghai, China. *Ecol. Indic.* 45, 332–339.
- Su, S., Zhou, X., Wan, C., Li, Y., Kong, W., 2016. Land use changes to cash crop plantations: crop types, multilevel determinants and policy implications. *Land Use Policy* 50, 379–389.
- Su, S., Xiao, R., Jiang, Z., Zhang, Y., 2012. Characterizing landscape pattern and ecosystem service value changes for urbanization impacts at an eco-regional scale. *Appl. Geogr.* 34, 295–305.
- Su, S., Liu, W., Wan, C., Li, J., Jin, X., Pi, J., Zhang, Q., Weng, M., 2017a. Economic benefit and ecological cost of enlarging tea cultivation in subtropical China: characterizing the trade-off for policy implications. *Land Use Policy* 66, 183–195.
- Su, S., Sun, Y., Lei, C., Weng, M., Cai, Z., 2017b. Reorienting paradoxical land use policies towards coherence: a self-adaptive ensemble learning geo-simulation of tea expansion under different scenarios in subtropical China. *Land Use Policy* 67, 415–425.
- Tilman, D., Socolow, R., Foley, J.A., Hill, J., Larson, E., Lynd, L., Pacala, S., Reilly, J., Searchinger, T., Somerville, C., Williams, R., 2009. Beneficial biofuels—the food, energy, and environment trilemma. *Science* 325, 4–5.
- Uuemaa, E., Mander, U., Mander, R., 2013. Trends in the use of landscape spatial metrics as landscape indicators: a review. *Ecol. Indic.* 28, 100–106.
- Weng, Y., 2007. Spatiotemporal changes of landscape pattern in response to urbanization. *Landscape Urban Plan.* 81, 341–353.
- Wainger, L.A., King, D.M., Mack, R.N., Price, E.W., Maslin, T., 2010. Can the concept of ecosystem services be practically applied to improve natural resource management decisions? *Ecol. Econ.* 69, 978–987.
- World Bank, 2010. *Environmental, Economic and Social Impacts of Oil Palm in Indonesia: A Synthesis of Opportunities and Challenges*. World Bank, Washington, D.C., USA.
- Wu, K., Ye, X., Qi, Z., Zhang, H., 2013. Impacts of land use/land cover change and socioeconomic development on regional ecosystem services: the case of fast-growing Hangzhou metropolitan area, China. *Cities* 31, 276–284.
- Xiao, R., Su, S., Mai, G., Zhang, Z., Yang, C., 2015. Quantifying determinants of cash crop expansion and their relative effects using logistic regression modeling and variance partitioning. *Int. J. Appl. Earth Observ. Geoinform.* 34, 258–263.
- Yi, Z., Cannon, C.H., Chen, J., Ye, C., Swetnam, R.D., 2014a. Developing indicators of economic value and biodiversity loss for rubber plantations in Xishuangbanna, southwest China: a case study from Menglung township. *Ecol. Indic.* 36, 788–797.
- Yi, Z., Wong, G., Cannon, C.H., Xu, J., Beckschäfer, P., Swetnam, R.D., 2014b. Can carbon-trading schemes help to protect China's most diverse forest ecosystems? A case study from Xishuangbanna, Yunnan. *Land Use Policy* 38, 646–656.
- Zhai, D., Cannon, C.H., Slik, J.W.F., Zhang, C., Dai, Z., 2012. Rubber and pulp plantations represent a double threat to Hainan's natural tropical forests. *J. Environ. Manage.* 96, 64–73.
- Zhang, L., Kono, Y., Kobayashi, S., 2014. The process of expansion in commercial banana cropping in tropical China: a case study at a Dai village, Mengla County. *Agric. Syst.* 124, 32–38.
- Zhang, Y., 2012. *Spatiotemporal Changes of Ecosystem Service Values of Agricultural Land*. Sichuan Agricultural University Ph.D thesis (In Chinese with English abstract).
- Ziegler, A.D., Fox, J.M., Xu, J., 2009. The rubber juggernaut. *Science* 324, 1024–1025.